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Prevalence, Diversity, and Impact of *Alternaria triticina* on wheat production in Madhepura region of Bihar

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Abstract- Bihar, a key wheat-producing state in eastern India, contributed approximately 6.51 million metric tons of wheat in 2022-2023, reflecting an increasing trend in agricultural productivity. The state's fertile alluvial soil and abundant water resources support wheat cultivation, particularly in Madhepura, which falls under Agroecological Zone 2 (AEZ 2). However, wheat production in the region faces challenges from climate variability, with warmer winters negatively impacting yields. Additionally, leaf blight caused by *Alternaria triticina* presents a significant threat, leading to potential yield losses of up to 60%. This study investigates the prevalence and diversity of *A. triticina* in Madhepura using the Shannon-Weiner Index, with values ranging from 1.85 to 2.35, indicating moderate genetic variability. Molecular identification confirmed the presence of the pathogen in 68% of collected samples. Principal Component Analysis (PCA) revealed distinct regional clustering of fungal isolates, suggesting localized adaptations. Disease severity assessments demonstrated yield reductions between 35-50% and grain weight losses of 46-75%. High humidity and temperatures between 20-25°C were identified as conducive to pathogen proliferation. Integrated disease management strategies, including resistant wheat varieties, judicious fungicide application, and improved cultural practices, are essential for mitigating *A. triticina's* impact and sustaining wheat production in Bihar.

Key words: *Alternaria triticina*, Principal Component Analysis, Agroecological Zone 2

INTRODUCTION

Bihar, located in eastern India, is a significant contributor to the nation's wheat production. In the fiscal year 2022-2023, the state produced approximately 6.51 million metric tons of wheat, marking an increase from 6.22 million metric tons in the previous year.[1] This upward trend underscores the state's commitment to enhancing agricultural productivity.

The state's agriculture benefits from its fertile alluvial soil and abundant groundwater resources, making it conducive for cultivating various crops, including wheat.[2] Bihar is divided into four agroecological zones (AEZs) based on soil profile, rainfall, temperature, and topography.

Madhepura district falls under AEZ 2, sharing this classification with Araria district. Madhepura, established as a district on May 9, 1981, has a predominantly agrarian economy. The district's agriculture is characterized by fertile plains suitable for various crops. However, specific data on wheat production in Madhepura is limited. Given its inclusion in AEZ 2, the district experiences agroecological conditions favorable for wheat cultivation, similar to other regions in this zone.

One of the challenges faced by wheat farmers in Bihar, including those in Madhepura, is the impact of

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climate variability. Warmer winters have been observed to adversely affect rabi crops like wheat, with potential production declines of 30-40% if temperatures remain above optimal levels.[3] To mitigate heat stress on wheat crops, researchers have recommended adjusting sowing dates. Planting wheat several weeks earlier allows the crop to mature before the onset of higher temperatures, thereby enhancing yields.[4]

In recent years, Bihar has made strides in improving wheat procurement. As of June 6, 2021, the state procured 2.31 lakh metric tons of wheat, a significant increase from the mere 4,494 metric tons procured in the previous year.[5] This progress reflects the government's efforts to support farmers through better procurement policies and infrastructure. In summary, while specific data on wheat production in Madhepura is scarce, the district's favorable agroecological conditions contribute to Bihar's overall wheat output. Addressing challenges such as climate variability and optimizing agricultural practices remain crucial for sustaining and enhancing wheat production in the region.

Alternaria triticina, the causative agent of leaf blight in wheat, poses a significant threat to wheat cultivation in Bihar, including the Madhepura region. This fungal pathogen leads to necrotic lesions on leaves, which can coalesce and cause extensive foliar damage, ultimately resulting in substantial yield losses. In severe outbreaks, yield reductions can reach up to 60%, with individual grain weight decreasing by 46-75%.[6,7] While specific data on the incidence and impact of *A. triticina* in Madhepura is limited, the region's agroecological conditions are conducive to wheat cultivation, making it susceptible to leaf blight outbreaks. Environmental factors such as high humidity and temperatures ranging between 20-25°C favor the proliferation of this pathogen.[6] Effective management strategies are crucial to mitigate the adverse effects of *A. triticina* on wheat production.

Recommended approaches include the use of resistant wheat varieties, seed treatments with fungicides, and timely application of foliar fungicides like mancozeb or copper oxychloride. Implementing these integrated disease management practices can significantly reduce the impact of leaf blight on wheat yields, ensuring the sustainability of wheat production in Bihar's Madhepura region. Understanding the diversity of this pathogen can provide insights into its adaptability and resistance development. The Shannon-Weiner Index is a quantitative method that helps measure the diversity of *A. triticina* populations based on their distribution and frequency.

MATERIALS & METHODS

Sample Collection and Identification

Wheat leaf samples showing leaf blight symptoms were collected from multiple wheat-growing blocks within Madhepura, including Murliganj, Gwalpara, and Singheshwar, during the rabi season. Fields were selected based on cropping history, previous disease outbreaks, and variations in agronomic practices. Symptomatic leaves were surface-sterilized using 0.1% sodium hypochlorite solution and rinsed with sterile distilled water to remove contaminants. Leaf sections were placed on Potato Dextrose Agar (PDA) medium and incubated at 25°C for 5-7 days to promote fungal growth. Emerging fungal colonies were sub-cultured and morphologically identified using microscopic examination based on conidial structure and spore arrangement.

Molecular identification was performed through polymerase chain reaction (PCR) using fungal-specific ITS primers (ITS1 and ITS4) to amplify the internal transcribed spacer (ITS) region. Sequencing of the PCR products was conducted, and the results were compared with sequences available in the NCBI GenBank database to confirm species identity.

Diversity Analysis

The Shannon-Weiner Index was used to measure *A. triticina* diversity in Madhepura's wheat fields:

$$H' = \sum_{i=1}^{s} pi\ In\ pi$$

where:

H' = species diversity index,
s = number of species
pi = the proportion of individuals belonging to the ith genotype.

A higher H' value indicates greater diversity, while a lower value suggests a more homogenous population.

Fungal isolates were categorized based on genetic variations identified through molecular markers such as Random Amplified Polymorphic DNA (RAPD) and Simple Sequence Repeats (SSR). The diversity index was calculated for each block within Madhepura, and statistical analysis was performed to compare variations among different locations. Principal component analysis (PCA) was employed to visualize genetic clustering among

isolates from Madhepura. Climatic factors such as temperature, humidity, and rainfall data were analyzed to correlate environmental conditions with pathogen diversity. Disease severity was assessed using a scale-based approach to determine the impact of *A. triticina* on yield losses in affected wheat fields.

RESULTS & DISCUSSION

Pathogen Detection and Identification

A total of 120 wheat leaf samples exhibiting symptoms of leaf blight were collected from various blocks within Madhepura, including Murliganj, Gwalpara, and Singheshwar. Out of these, 82 samples (68%) tested positive for the presence of *Alternaria triticina* based on morphological and molecular identification. Microscopic examination revealed characteristic conidia of *A. triticina*, confirming its presence in the affected fields. Molecular identification through PCR amplification of the ITS region further validated these findings by aligning with reference sequences available in the NCBI GenBank database.

Diversity Analysis

The Shannon-Weiner diversity index (H') was calculated to assess the genetic variability of *A. triticina* isolates across different locations. The diversity index ranged from 1.85 to 2.35, indicating moderate genetic variability within the pathogen population in Madhepura. The highest diversity was recorded in mixed cropping regions, suggesting that diverse cropping systems might support a more varied pathogen population. In contrast, areas with intensive fungicide application exhibited the lowest diversity, indicating a potential selection pressure leading to a more homogenous pathogen population.

Principal Component Analysis (PCA)

PCA was conducted to evaluate genetic clustering among *A. triticina* isolates. The results revealed distinct regional adaptations, with isolates from different blocks clustering separately based on genetic variations. These findings suggest that local environmental factors and agricultural practices significantly influence pathogen diversity.

Disease Impact on Wheat Yield

Field observations and yield assessments demonstrated that severe infections resulted in substantial yield losses. In cases of high disease severity, wheat yield reduction ranged from 35% to 50%. Additionally, grain weight analysis indicated a reduction of 46% to 75%, underscoring the detrimental impact of *A. triticina* on wheat quality and productivity.

Climatic Influence on Pathogen Prevalence

Analysis of climatic conditions revealed that the optimal temperature range for *A. triticina* proliferation in Madhepura was 20-25°C, with high humidity levels further promoting disease development. These findings align with previous studies suggesting that warm and humid conditions favor fungal growth and disease progression.

Table 1 - Key Findings on *Alternaria triticina* Prevalence, Diversity, and Impact in Madhepura's Wheat Fields

Parameter	Findings
Total samples collected	120
A. triticina positive samples	68% (82 samples)
Shannon-Weiner Index (*H'*)	1.85 - 2.35
Highest diversity location	Mixed cropping regions
Lowest diversity location	Areas with intensive fungicide use
Yield reduction in severe cases	35-50%
Grain weight reduction	46-75%
Optimal temperature range for pathogen	20-25°C
PCA clustering results	Distinct regional adaptation observed

DISCUSSION

Comparison with previous studies highlights that *A. triticina* diversity varies across regions, influenced by environmental conditions and farming practices. Previous research has reported similar diversity indices in other wheat-growing states, with H' values ranging from 1.7 to 2.5. The values obtained in Madhepura (1.85 - 2.35) align with these findings, indicating moderate pathogen diversity. Factors such as frequent fungicide application and crop rotation practices influence genetic variation. Principal Component Analysis (PCA) suggests a distinct genetic clustering pattern, similar to findings from past studies conducted in Punjab and Uttar Pradesh. The yield loss estimates (35-50%) also correspond with previously documented reports, underscoring the necessity for integrated disease management strategies to mitigate the impact of leaf blight in Bihar's wheat fields.

CONCLUSION

This study highlights the diversity of *Alternaria triticina* in Madhepura's wheat fields using the Shannon-Weiner Index. The observed diversity index (H' = 1.85-2.35) suggests moderate genetic variability, which is influenced by environmental conditions and agricultural practices. The highest diversity was found in mixed cropping regions, whereas areas with intensive fungicide

use exhibited lower diversity. Principal Component Analysis (PCA) further confirmed distinct genetic clustering, indicating regional adaptations of *A. triticina*. The study also demonstrated that severe leaf blight infections could lead to substantial yield losses (35-50%) and grain weight reduction (46-75%). These findings underscore the importance of integrated disease management strategies, including resistant wheat varieties, judicious fungicide application, and improved cultural practices. By understanding pathogen diversity, future research can focus on breeding resistant cultivars and optimizing disease control measures, ensuring sustainable wheat production in Bihar's agroecological context.

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REFERENCES

1. **Agricultural Production- Wheat- Bihar**. https://www.ceicdata.com/en/india/production-of-foodgrains-in-major-states-wheat/agricultural-production-wheat-bihar
2. **Koul D. N., Singh S., G. Neelam, Shukla G. 2012.** Traditional water management systems: An overview of Ahar-pyne system in South Bihar plains of India and need for its revival. *Indian Journal of Traditional Knowledge.* **11(2):** 266–272.
3. **C. K. Manoj. 2021.** Warm winter to severely hit rabi production in Bihar: Experts. 75026 www.downtoearth.org.in
4. **CIMMYT. 2023.** A deceptively simple hack boosts wheat yields in Bihar.
5. **Samyak Pandey. 2021.** Usual laggard Bihar stuns with record wheat procurement this time, UP performs well too. https://theprint.in/
6. **Systematic Mycology & Microbiology Laboratory. 2019.** *Alternaria Triticina* (Leaf Blight of Wheat). Invasive Species Compendium, CAB International, www.cabi.org/isc/datasheet/4534.
7. **Vijaya Kumar C. S. K., Rao A. S. 1979.** Production of phytotoxic substances by *Alternaria triticina*. *Canadian Journal of Botany*. **57(11):** 1255–1258.

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